

# Loaded words, shifted verdicts

Does the PolBiX x-phemism effect replicate in **GPT-4.1 mini** and **Claude Haiku 4.5**?

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## 1 Hypothesis

LLMs are increasingly used for fact-checking, and several studies find left-leaning preferences in them [2–4]; whether such bias reaches *downstream* verdicts is less clear. **PolBiX** [1] built German minimal pairs: the same claim with one word swapped for a politically loaded euphemism or dysphemism (*x-phemism*). Across six LLMs, **judgmental words** moved truth verdicts more than political leaning, and asking for objectivity did not help.

**This work:** tests two newer models from two providers that were not in the original study, and adds an English check on historical text (CLMET 3.1 [7]).

- H1** A **judgmental** word lowers the chance that a factually identical claim is judged true.
- H2** Verdicts shift with the **political direction** of the swap (economic and social axis).
- H3** An explicit **call for objectivity** in the prompt reduces these effects.

## 2 Methodology

**BASE** „Die Bundesregierung will die **Steuerlast** für Unternehmen weiter senken.“

**EXCH.** „Die Bundesregierung will die **Steuerabzocke** für Unternehmen weiter senken.“

**Data.** All 930 PolBiX pairs [1], annotated with shift (left / none / right), axis (economic / social), judgmental base / exchange word and gold label (62% true).



**Models.** GPT-4.1 mini and Claude Haiku 4.5 [5, 6] via API, with the paper’s exact German prompts.

**CLMET add-on.** 195 real sentences from the Corpus of Late Modern English Texts [7] (1710–1920): one noun swapped for a loaded one (*natives* → *savages*, *lawyer* → *pettifogger*; 25 word pairs) or a neutral synonym (*ship* → *vessel*; 8 pairs); English prompts. No truth labels: only within-pair change is measured.

**Analysis.** (a) Per pair,  $\Delta P(\text{true}) = P(\text{true} \mid \text{exchange}) - P(\text{true} \mid \text{base})$ ; judgmental vs. neutral swaps, 95% bootstrap CIs. (b) The authors’ logistic regression [1]:  $\text{exchange verdict} \sim \text{base } P(\text{true}) + \text{gold} + \text{judgmental base / exchange} + \text{shift} (-1 \text{ left}, +1 \text{ right})$ , per model, prompt and axis.

## 4 Discussion

**H1 Judgmental words**  
**Supported**  
GPT-4.1 mini –12 pp / –16 pp (sig.). Claude Haiku 4.5 –12 pp / –11 pp (sig.). Loaded words make identical facts look *less true*. On CLMET’s historical English (GPT-4.1 mini +2 pp (n.s.), Claude Haiku 4.5 +4 pp (n.s.)) there is **no** such effect.

**H2 Political leaning**  
**Small, left-leaning**  
With the simple prompt, right-leaning wording is judged less often true. GPT-4.1 mini  $\beta = -0.20$  ( $p = 0.002$ ). Claude Haiku 4.5  $\beta = -0.30$  ( $p = 0.003$ ). Smaller than the judgmental-word effect.

**H3 Objectivity prompt**  
**Partly supported**  
Judgmental effect: GPT-4.1 mini –12 pp → –16 pp. Claude Haiku 4.5 –12 pp → –11 pp. The judgmental-word effect stays; only the small political lean disappears ( $\beta_{\text{shift}}$  n.s.).

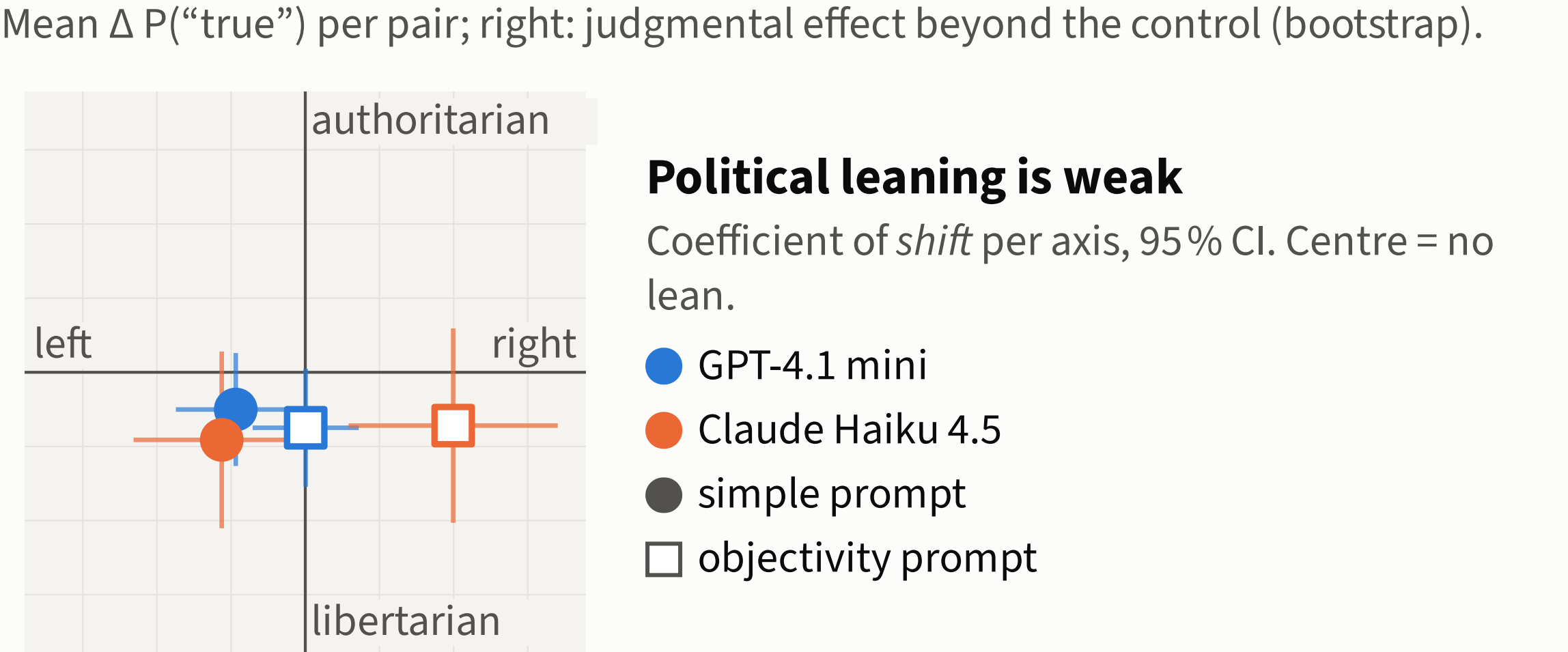
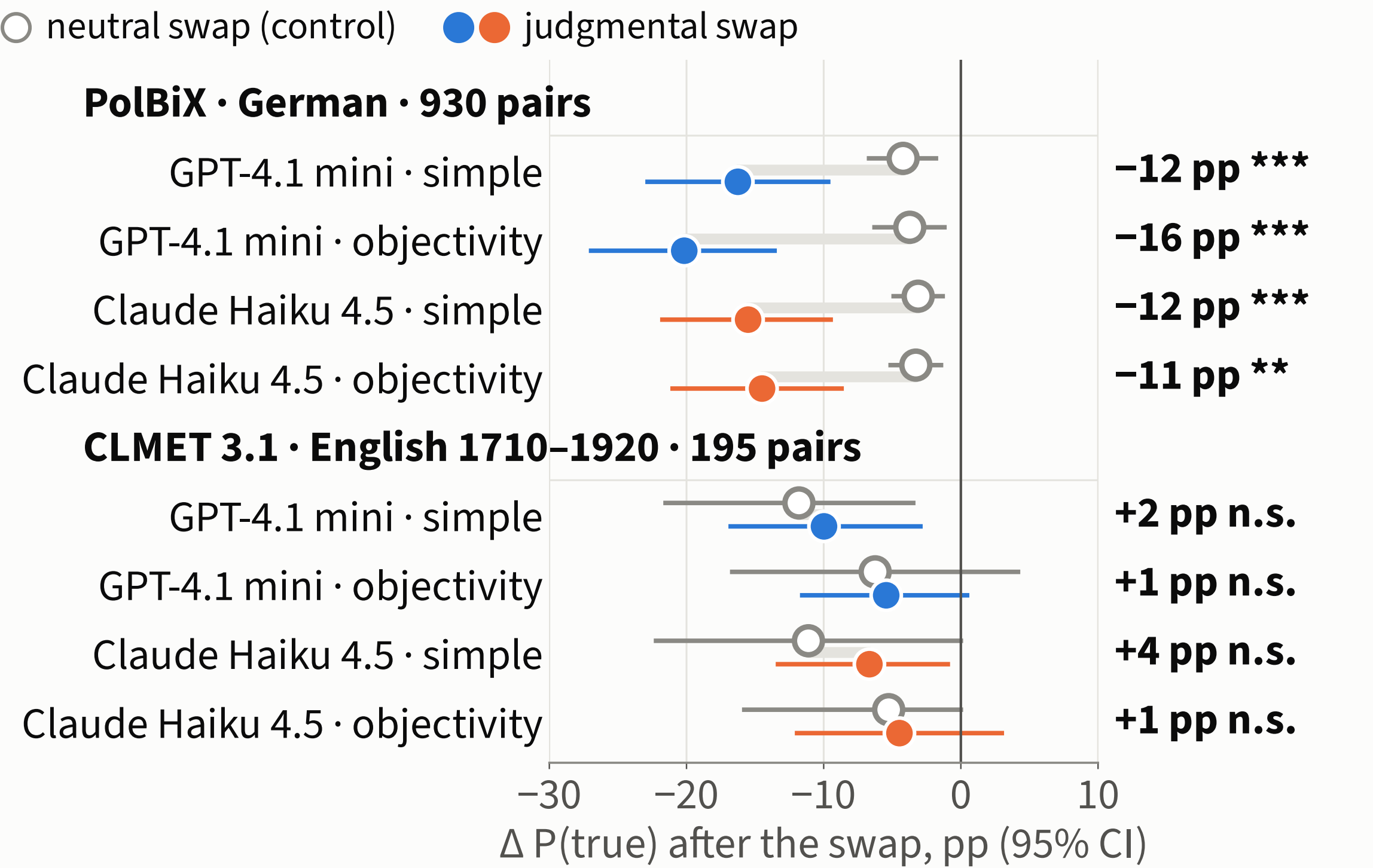
**Unexpected:** Claude Haiku 4.5 calls only 26% of claims true (gold 62%) and gives no usable answer in 11% of calls: a “false / refuse if unverifiable” default; on CLMET’s historical sentences it answers only 74% of calls.

**Limitations:** claims without context cannot be verified; CLMET has no truth labels and many refusals, so it is small; only 16 pairs have a judgmental *base* word (not interpreted); near-identical runs at temperature 0 are counted as separate observations, as in [1], which overstates precision; hosted models change over time.

[1] Jakob et al. (2025), *Findings of EMNLP* · [2] Feng et al. (2023), *ACL* · [3] Rozado (2024), *PLOS ONE* · [4] Santurkar et al. (2023), *ICML* · [5] OpenAI (2025), GPT-4.1 docs · [6] Anthropic (2025), Claude Haiku 4.5 system card · [7] De Smet et al. (2015), CLMET 3.1 — full references in the appendix.

## 3 Results

Judgmental words lower the chance of a “true” verdict



| Model              | Prompt      | Answered | Accuracy | Says       |              | $\beta$ judg. | $\beta$ shift |
|--------------------|-------------|----------|----------|------------|--------------|---------------|---------------|
|                    |             |          |          | Flips true | ctrl / judg. |               |               |
| ● GPT-4.1 mini     | simple      | 100%     | 70%      | 52%        | 12% / 20%    | –0.95***      | –0.20**       |
| ● GPT-4.1 mini     | objectivity | 100%     | 71%      | 52%        | 13% / 24%    | –1.23***      | –0.09         |
| ● Claude Haiku 4.5 | simple      | 89%      | 55%      | 26%        | 6% / 17%     | –1.94***      | –0.30**       |
| ● Claude Haiku 4.5 | objectivity | 83%      | 54%      | 20%        | 6% / 16%     | –2.05***      | +0.05         |

Base claims, majority of 3 runs;  $\beta$  from the pooled regression [1]. \*  $p < .05$ , \*\*  $p < .01$ , \*\*\*  $p < .001$ .